THE ASSAM
ROYAL GLOBAL UNIVERSITY
GUWAHATI

Royal School of Information Technology (RSIT)

Course Structure & Syllabus

(Based on National Education Policy 2020)

For

Bachelor of Computer Application (BCA)

W.E.F

AY: 2026-2027

Course Structure and Syllabus of the Framework

1st Semester							
Sl. No.	**Subject Code**	**Names of subjects**	**Level of Course**	**Credit**	**L**	**T**	**P**
Core							
1	CAP052C101	Mathematical Foundations to Computer Science	100	4	3	0	0
2	CAP052C102	Introduction to C Programming	100	3	3	0	0
3	CAP052C103	Introduction to Fundamental Electronics and Digital Logic	100	3	3	0	0
4	CAP052C112	Introduction to C Programming Lab	100	1	3	0	2
Interdisciplinary							
6	IKS992K101	Introduction to Indian Knowledge System-I	100	3	3	0	0
Ability Enhancement Courses (AEC)							
7	AEC982A101	Communicative English-I	100	2	2	0	0
Skill Enhancement Courses (SEC)							
8	CAP052S101	Computer Hardware and Networking	100	3	3	0	0
Value Addition Courses (VAC)							
9	CAP052V101	Office Automation	100	3	3	0	0
MOOCS							
10	MOOCS	**One 8-week Course from SWAYAM /MOOCS as per the direction of the Department**	100	3 or 4	0	0	0
		TOTAL		**22**			
2nd Semester							
Sl. No.	**Subject Code**	**Names of subjects**	**Level of Course**	**Credit**	**L**	**T**	**P**
Core							
1	CAP052C201	Data Structures	100	3	3	0	0
2	CAP052C202	Computer Architecture	100	3	3	0	0
3	CAP052C203	OOP using C++	100	3	3	0	0
4	CAP052C211	Data Structures Lab	100	1	0	0	2
5	CAP052C213	OOP using C++ Lab	100	1	0	0	2
Interdisciplinary							
6	IKS992K201	Introduction to Indian Knowledge System-II	100	3	3	0	0
Ability Enhancement Courses (AEC)							
7	AEC982A201	Communicative English	100	2	2	0	0
Skill Enhancement Courses (SEC)							
8	CAP052S201	Internet Technologies	100	3	3	0	0
Value Addition Courses (VAC)							
9	CAP052V201	Introduction to Data Analytics	100	3	3	0	0
MOOCS							
10	MOOCS	**One 8-week Course from SWAYAM /MOOCS as per the direction of the Department**	100	3 or 4	0	0	0
		TOTAL		**22**			
3rd Semester							
Sl. No.	**Subject Code**	**Names of subjects**	**Level of Course**	**Credit**	**L**	**T**	**P**
Core							

1	CAP052C301	Python Programming	200	3	3	0	0
2	CAP052C302	Database Management Systems	200	3	3	0	0
3	CAP052C303	Introduction to Artificial Intelligence	200	4	4	0	0
4	CAP052C311	Python Programming Lab	200	1	0	0	2
5	CAP052C312	Database Management Systems Lab	200	1	0	0	2
Interdisciplinary							
6	CAP052I301	IDC-I (E- Commerece and Digital Marketing)	200	3	0	0	3
Ability Enhancement Courses (AEC)							
7	CEN982A301	Communicative English III	200	2	2	0	0
Skill Enhancement Courses (SEC)							
8	CAP052S301	System Administration	200	3	3	0	0
MOOCS							
9	MOOCS	**One 8-week Course from SWAYAM /MOOCS as per the direction of the Department**	200	3 or 4	0	0	0
		TOTAL		**20**			
4th Semester							
Sl. No.	**Subject Code**	**Names of subjects**	**Level of Course**	**Credit**	**L**	**T**	**P**
Core							
1	CAP052C401	Operating System Concepts	200	3	3	0	0
2	CAP052C402	Introduction to Data Communication Networks	200	3	3	0	0
3	CAP052C403	Design and Analysis of Algorithms	200	3	3	0	0
4	CAP052C404	Introduction to Machine Learning	200	3	3	0	0
5	CAP052C405	Software Engineering	200	3	3	0	0
6	CAP052C411	Operating System Concepts	200	1	0	0	2
7	CAP052C412	Introduction to Data Communication Networks Lab	200	1	0	0	2
8	CAP052C414	Introduction to Machine Learning Lab	200	1	0	0	2
Ability Enhancement Courses (AEC)							
9	AEC982A101	Communicative English	200	2	2	0	0
MOOCS							
10	MOOCS	**One 8-week Course from SWAYAM /MOOCS as per the direction of the Department**	200	03-Apr	0	0	0
		TOTAL		**20**			
5th Semester							
Sl. No.	**Subject Code**	**Names of subjects**	**Level of Course**	**Credit**	**L**	**T**	**P**
Core							
1	CAP052C501	Full Stack Development	300	4	4	0	0
2	CAP052C502	Cryptography and Network Security	300	4	3	0	1
3	CAP052C503	Introduction to Deep Learning	300	4	3	0	1
Internship							
4	CAP052C524	Summer Internship	300	4	0	0	0
Project							
5	CAP052C525	Project Phase-I	300	4	0	0	0
MOOCS							
6		**One 8-Week Course from SWAYAM /MOOCS as per the direction of the Department**	300	3 or 4	0	0	0
		TOTAL		**20**			
6th Semester							

Sl. No.	Subject Code	Names of subjects	Level of Course	Credit	L	T	P
Core							
1		Compiler Design/ Mobile Application Development	300	4	3	0	1
2		Introduction to Data Science	300	4	3	0	1
3		Introduction to Big Data Analytics/Cloud Computing	300	4	4	**0**	0
Project							
4		Project Phase-II	300	8	0	0	0
MOOCS							
5	MOOCS	**One 8-Week Course from SWAYAM /MOOCS as per the direction of the Department**	300	03-Apr	0	0	0
		TOTAL		**20**			
7th Semester							
Sl. No.	**Subject Code**	**Names of subjects (Suggested)**	**Level of Course**	**Credit**	L	**T**	**P**
Core							
1		Introduction to Data Warehousing	400	4	4	0	0
2		Introduction to Natural Language Processing	400	3	3	0	0
3		Mobile Computing	400	4	4	0	0
4		Introduction to Natural Language Processing Lab	400	1	0	0	2
Dissertation							
5		Dissertation-I	400	8	0	0	0
MOOCS							
6	MOOCS	**One 8-week Course from SWAYAM /MOOCS as per the direction of the Department**	400	03-Apr	0	0	0
		TOTAL		**20**			
8th Semester							
Sl. No.	**Subject Code**	**Names of subjects (Suggested)**	**Level of Course**	**Credit**	**L**	**T**	**P**
Major							
1		Intelligent Systems and Explainable AI	400	4	4	0	0
2		Speech Processing	400	4	4		0
Dissertation							
3	MOOCS	Dissertation	400	12	0	0	0
MOOCS							
4		One 8-week Course from SWAYAM /MOOCS as per the direction of the Department	400	03-Apr			
Advanced Level Core Course instead of Dissertation							
5	CAP052M802	Prompt Engineering	400	4	3	1	0
6	CAP052M803	Blockchain Technologies/ Social Network Analysis	400	4	3	1	0
7	CAP052M804	Quantum Computing/ Bioinformatics	400	4	3	1	0
		Total		**20**			

Detailed Syllabus of all the Courses Semester-wise

SYLLABUS (1st SEMESTER)

Paper I/Subject Name: Discrete Structures		**Subject Code: CAP052M101**
Course Type: Core		**Course Level:100**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Course Objective:

The objectives of the course are to provide the students with a basic understanding of fundamental mathematical concepts such as sets, functions, matrix algebra, discrete mathematics etc and also to enable the students to use mathematical models and techniques to analyze and understand problems in computer science.

Prerequisites: None

Course Outcomes

On successful completion of the course, the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** and **explain** the fundamental concepts of sets, relations, functions, and propositional logic used in discrete mathematical structures.	**BT 1 & 2**
CO 2	**Apply** counting techniques, recurrence relations, matrices, and logical methods to solve computational and mathematical problems	**BT 3**
CO 3	**Analyze** properties of relations, functions, matrices, and logical propositions to establish correctness and validity of mathematical models.	**BT 4**
CO 4	**Evaluate** logical arguments, matrix solutions, and proof strategies using rules of inference and mathematical induction.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Sets, Relations and Functions**	Set, Set Operations, Properties of Set operations, Subset, Venn Diagrams, Cartesian Products, Relations on a Set, Properties of Relations, Representing Relations using matrices and digraphs, Types of Relations, Equivalence Relation, Equivalence relation and partition on set, Closures of Relations, Warshall's algorithm, Functions, properties of functions (domain, range), composition of functions, surjective (onto), injective (one-to-one) and bijective functions, inverse of functions. Some useful functions for Computer Science: Exponential and Logarithmic functions, Polynomial functions, Ceiling and Floor functions.	**15**
II	**Counting and Recurrence Relations**	Basics of counting, Pigeonhole principle, permutation, combination, Binomial coefficients, Binomial theorem. Recurrence relations, modelling recurrence relations with examples, like Fibonacci numbers, the tower of Hanoi problem. Solving linear recurrence relation with constant coefficients using characteristic equation roots method. Pigeon-hole principle, permutation and combination, Recurrence relations, Generating functions.	**15**
III	**Matrix Algebra**	Types of matrices, algebra of matrices–addition, subtraction, and multiplication of matrices, determinant of a matrix, symmetric and skew-symmetric matrices, orthogonal matrix, rank of a matrix, inverse of a matrix, applications of matrices to solve system of linear equations, Eigen values and Eigen vectors, Caley-Hamilton theorem.	**15**

IV	**Propositional Logic**	Logic and Methods of Proofs: Propositions, logical operations (basic connectives), compound statements, construction of truth table, quantifiers, conditional statements, tautology, contradiction, contingency, logical equivalence. Conjunctive Normal Forms (CNF) and Disjunctive Normal Forms (DNF). Methods of proofs: Rules of inference for propositional logic, modus ponens, modus tollens, syllogism, proof by contradiction, Mathematical Induction.	**15**
Total			**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	-	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. *A text book of Discrete Mathematics*, Sarkar S. K., Revised Edition, 2016, S Chand & Co Ltd.
2. *Discrete Mathematics and Its Applications,* Kenneth H. Rosen, 8th Edition, 2019, McGraw-Hill Education.

Reference Books:

1. Deo N; *Graph Theory with applications to engineering and computer science*, New Edition, 2009, PHI Learning Private Limited.
2. Chandrasekaran N. and Umaparvathi, *Discrete Mathematics*, Eastern Economic Edition, 2013, PHI
3. *Discrete Mathematics and its Applications,* Rosen, K.H., 6th Edition, 2006, McGraw Hill.
4. Tremblay, J.P. and Manohar, R., *Discrete Mathematical Structures with Applications to Computer Science*, 35th Reprint, 2007, Tata McGraw Hill

Paper II/Subject Name: Introduction to C Programming		**Subject Code: CAP052M102**
Course Type: Core		**Course Level:100**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to give the students exposure to computer programming and make them capable of using the concepts to solve basic as well as advanced computing problems.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** different constructs and building blocks present in C programming Language	**BT 1**
CO 2	**Demonstrate** the working of C programming language and its control structures.	**BT 2**

CO 3	**Apply** the programming concepts to solve various problems.	**BT 3**
CO 4	**Analyse** and debug the errors while writing the programs.	**BT 4**
CO 5	**Assess** and **design** a new algorithm to solve a new real-life problem	**BT 5**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**C Programming Fundamentals**	History and importance of C language, Basic structure of programs, programming style, execution of C programs. Character set, Tokens, Keywords and Identifiers, Constants, variables, data types, statements, comments, declaration of storage class, assigning values to variables. Basic idea of Computer Algorithms and Flow Charts. Managing I/O, reading and writing characters, formatted Input/output. Arithmetic operators, relational operators, logical operators, assignment operators, increment & decrement operators, conditional operators, bitwise operators, special operators. Arithmetic expressions, operator precedence & associativity.	**15**
II	**Decision Making, Branching & Lopping**	Importance of decision making, decision making with *if* statement, *if-else* statement, nested *if-else* statements, *switch-case* statement, *goto* statement, the *?:* operator, examples. Importance of loping, the *while* statement, *do-while* statement, *for* statement, nested looping, examples.	**15**
III	**Arrays, Strings & User-Defined Functions**	Significance of Arrays, creation and use of one- & two-dimensional arrays, Dynamic arrays. Declaration and use of string variables, reading and writing strings, operations on strings. Benefits of user-defined functions, creation and use of user-defined functions, parameter passing, return types.	**15**
IV	**Advanced Programming Concepts**	Creation and use of Structures and Unions in programs. Introduction to Pointers, declaration & initialization of pointer variables, accessing a variable through its pointer. Defining, opening & closing files in C, Input/output operations on files.	**15**
Total			**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 20 NCH = 60 NCH	--	30 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Computer Fundamentals and Programming in C,* Reema Thareja, 2nd Edition, 2016, Oxford University Press, Delhi.

Reference Books:

1. E Balaguruswamy, *Computing Fundamentals and C Programming*, 1st Edition, 2017, McGraw Hill.
2. Venugopal and Prasad, *Mastering C*, 2nd Edition, 2017, Tata McGraw Hill.
3. Yashawant Kanetkar, *Let us C,* 15th Edition, 2017, BPB Publication.

Paper III/Subject Name: Introduction to Fundamental Electronics and Digital Logic Subject Code:		
Course Type: Core		**Course Level:100**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to introduce the students with the basic concepts of electrical and electronic components used in digital systems, explain the working principles of semiconductor devices and digital electronic circuits, develop an understanding of number systems, Boolean algebra, and logic gates and enable students to analyze and design simple combinational and sequential digital circuits.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** and **explain** the characteristics of basic electronic components and semiconductor devices used in digital systems.	**BT 1 & 2**
CO 2	**Apply** number systems, Boolean algebra, and logic gates to solve digital logic problems.	**BT 3**
CO 3	**Analyse** and simplify combinational logic circuits using Karnaugh Maps and standard logic components.	**BT 4**
CO 4	**Evaluate** digital circuit designs for correctness and efficiency.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Fundamentals of Electronics**	Introduction to Electronics and Digital Systems, Passive Components: Resistors, Capacitors, Inductors, Active Components: Diodes, LEDs, Zener Diode, Bipolar Junction Transistor (BJT), MOSFET (Introduction), Semiconductor Basics, Power Supply Concepts, Introduction to Integrated Circuits (ICs),	**15**
II	**Number Systems and Boolean Algebra**	Decimal, Binary, Octal and Hexadecimal Number Systems, Number System Conversions, Binary Arithmetic, Signed Number Representation, Complements (1's and 2's), Binary Codes, BCD, Gray Code, ASCII, Boolean Algebra, Boolean Laws and Theorems, De Morgan's Theorem, Simplification of Boolean Expressions	**15**
III	**Logic Gates**	Logic Gates: AND, OR, NOT, NAND, NOR, XOR, XNOR, Universal Gates, Logic Families (Introduction), Karnaugh Maps (up to four variables)	**15**
IV	**Combinational Logic Circuits**	Half Adder, Full Adder, Half Subtractor, Full Subtractor, Multiplexer, Demultiplexer, Encoder, Decoder, Comparator	**15**
Total			**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**

3*20 NCH = 60 NCH	-	30 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Project)

Text Book:

1. *Digital Logic and Computer Design,* M. Morris Mano, 6th Edition, 2018, Pearson Education.
2. *Modern Digital Electronics,* R. P. Jain, 5th Edition, 2018, McGraw-Hill Education.

Reference Books:

1. Anand Kumar, *Fundamentals of Digital Circuits*, 4th Edition, 2016, PHI Learning.

Paper IV/Subject Name: Introduction to C Programming Lab		**Subject Code: CAP052M112**
Course Type: Core		**Course Level:100**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to give the students exposure to computer programming and make them capable of using the concepts to solve basic as well as advanced computing problems.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:

SI No	Course Outcome	Blooms Taxonomy Level
CO 1	**Define** different constructs and building blocks present in C programming Language	**BT 1**
CO 2	**Demonstrate** the working of C programming language and its control structures.	**BT 2**
CO 3	**Apply** the programming concepts to solve various problems.	**BT 3**
CO 4	**Analyse** and debug the errors while writing the programs.	**BT 4**
CO 5	**Assess** and **design** a new algorithm to solve a new real-life problem	**BT 5**

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. Character set, Tokens, Keywords and Identifiers, Constants, variables, data types, statements, comments, declaration of storage class, assigning values to variables.
2. Managing I/O, reading and writing characters, formatted Input/output.
3. Arithmetic operators, relational operators, logical operators, assignment operators, increment & decrement operators, conditional operators, bitwise operators, special operators. Arithmetic expressions, operator precedence & associativity.
4. Importance of decision making, decision making with if statement, if-else statement, nested if-else statements, switch-case statement, goto statement, the ?: operator.
5. Importance of loping, the while statement, do-while statement, for statement, nested looping.

6. Significance of Arrays, creation and use of one & two dimensional arrays, Dynamic arrays.
7. Declaration and use of string variables, reading and writing strings, operations on strings.
8. Benefits of user-defined functions, creation and use of user-defined functions, parameter passing, return types.
9. Creation and use of Structures and Unions in programs.
10. Use of Pointers, declaration & initialization of pointer variables, accessing a variable through its pointer.
11. Defining, opening & closing files in C, Input/output operations on files.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	1 * 20 NCH = 20 NCH	10 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Computer Fundamentals and Programming in C,* Reema Thareja, 2nd Edition, 2016, Oxford University Press, Delhi.

Reference Books:

1. E Balaguruswamy, *Computing Fundamentals and C Programming*, 1st Edition, 2017, McGraw Hill.
2. Venugopal and Prasad, *Mastering C*, 2nd Edition, 2017, Tata McGraw Hill.
3. Yashawant Kanetkar, *Let us C,* 15th Edition, 2017, BPB Publication.

Paper V/Subject Name: Digital Logic and Design Lab		**Subject Code:**
Course Type: Core		**Course Level:100**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to introduce the students to the fundamentals of digital systems and number representation, develop understanding of Boolean algebra and logic gate operations, enable students to design and analyze combinational and sequential digital circuits and to provide foundational knowledge required for computer organization and hardware systems.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** and **explain** number systems, Boolean algebra, and logic gates used in digital systems.	**BT 1 & 2**
CO 2	**Apply** Boolean laws and Karnaugh maps to simplify logic expressions.	**BT 3**
CO 3	**Analyse** the behaviour of various sequential circuits using flip-flops and counters.	**BT 4**
CO 4	**Evaluate** digital circuit designs for correctness and efficiency.	**BT 5**

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

1. Study of logic gates and verification of truth tables of AND, OR, NOT, NAND, NOR, XOR, and XNOR gates.
2. Number system conversion and binary arithmetic operations.
3. Verification of Boolean algebra laws and De Morgan's theorem.
4. Simplification of Boolean expressions using Karnaugh Maps.
5. Design and implementation of Half Adder and Full Adder circuits.
6. Design and implementation of Half Subtractor and Full Subtractor circuits.
7. Design and implementation of Multiplexer and Demultiplexer circuits.
8. Design and implementation of Encoder and Decoder circuits.
9. Study and implementation of SR, JK, D, and T Flip-Flops.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	1*20 NCH = 20 NCH	10 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Project)

Text Book:

1. *Digital Logic and Computer Design,* M. Morris Mano, 6th Edition, 2018, Pearson Education.
2. *Modern Digital Electronics,* R. P. Jain, 5th Edition, 2018, McGraw-Hill Education.

Reference Books:

1. Anand Kumar, *Fundamentals of Digital Circuits*, 4th Edition, 2016, PHI Learning.

Paper VI/Subject Name: Introduction to Indian Knowledge System-I		**Subject Code: IKS992K101**
Course Type: IKS		**Course Level: 100**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objectives:
This foundation course is designed to present an overall introduction to all the streams of IKS relevant to the UG programme. It would enable students to explore the most fundamental ideas that have shaped Indian Knowledge Traditions over the centuries.

Prerequisite: None

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Illustrate** literature of Indian civilization-the Vedic – Itihasas, languages, mathematics, and Ayurveda.	**BT 2**
CO 2	**Explain** observation of the motion of celestial bodies in the Vedic corpus	**BT 3**

Modules	**Topics**	**Course content**	**Periods**
I	**Bharatavarsha —A Land of Rare Natural**	Demographical features of the ancient Bharatvarsha, Largest cultivable area in the world. Protected and nurtured by Himalayas. The Sindhu-Ganga plain and the great coastal plains. The great	**10**

	Endowments	rivers of India. Climatic changes: Abundant rains, sunshine and warmth, vegetation, animals and mineral wealth. Most populous country in the world. India's prosperity held the world in thrall. Splendid geographical isolation of India and the uniqueness of Indian culture.	
II	**Foundational Literature of Indian Civilization:**	The Vedic Corpus. The Itihasas— Ramayana and Mahabharata, and their important regional versions. The Puranas. Foundational Texts of Indian Philosophies, including the Jaina and Bauddha. Foundational Texts of Indian Religious Sampradayas, from the Vedic period to the Bhakti traditions of different regions. i. **The Vedangas and Other Streams of Indian Knowledge System:** The Vedic Corpus: Introduction to Vedas and synopsis of the four Vedas and Sub-classification of Vedas; Messages in Vedas; Introduction to Vedāṅgas : Siksha, Vyakarana, Chandas, Nirukta, Jyotisha and Kalpa ; Vedic Life: Distinctive Features. Other streams of Indian Knowledge System such as Ayurveda, Sthapatya, Natyasastra, Dharmasastra, Arthasastra, etc. The Indian way of continuing the evolution of knowledge through commentaries, interpretations and revisions of the foundational texts. The large corpus of literature in Indian languages. **ii. Indian Language Sciences:** Language Sciences and the preservation of the Vedic corpus. Varnamala of Indian languages based on classification of sounds on the basis of their origin and effort involved. The special feature of the scripts of most Indian languages, that each symbol is associated with a unique sound. Word formation in Sanskrit and Indian languages. Major insights in the Science of Vyakarana as established by Panini. Important texts of Indian Language Sciences —Siksha or phonetics, Nirukta or etymology, Vyakarana or Grammar, Chandas or Prosody. Navyanyaya and Navya-vyakarana in Navadvipa, Varanasi and West and South India. iii. **Indian Mathematics:** Numbers, fractions and geometry in the Vedas. Decimal nomenclature of numbers in the Vedas. Zero and Infinity. Simple constructions from Sulba-sutras. The development of the decimal place value system which resulted in a simplification of all arithmetical operations. Linguistic representation of numbers. Important texts of Indian mathematics. Brief introduction to the development of algebra, trigonometry and calculus. How Indian mathematics continued to flourish in the 18/19/20th centuries. Kerala School. Ramanujan.	**24**
III	**Indian Astronomy**	Ancient records of the observation of the motion of celestial bodies in the Vedic corpus. Sun, Moon, Nakshatra & Graha. Astronomy as the science of determination of time, place and direction by observing the motion of the celestial bodies. The motion of the Sun and Moon. Motion of equinoxes and solstices. Elements of Indian calendar systems as followed in different regions of India. Important texts of Indian Astronomy. Basic ideas of the planetary model of Aryabhata and its revision by Nilakantha. Astronomical instruments. How Indian astronomy continued to flourish in the 18/19th centuries. Astronomical endeavours of Jaisingh, Sankaravarman, Chandrasekhara Samanta.	**15**
IV	**Indian Health Sciences**	Vedic foundations of Ayurveda. Ayurveda is concerned both with maintenance of good health and treatment of diseases. Basic concepts of Ayurveda. The three Gunas and Three Doshas, Pancha-mahabhuta and Sapta-dhatu. The importance of Agni (digestion). Six Rasas and their relation to Doshas. Ayurvedic view of the cause of diseases. Dinacharya or daily regimen for the maintenance of good health. Ritucharya or seasonal regimen. Important Texts of Ayurveda. Selected extracts from Astāngahrdaya (selections from Sūtrasthāna) and Suśruta-Samhitā (sections on plastic surgery,	**17**

		cataract surgery and anal fistula). The large pharmacopeia of Ayurveda. Charaka and Sushruta on the qualities of a Vaidya. The whole world is a teacher of the good Vaidya. Charaka's description of a hospital. Hospitals in ancient and medieval India. How Ayurveda continued to flourish till 18/19th centuries. Surgical practices, inoculation. Current revival of Ayurveda and Yoga.	
			Total **66**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 22 NCH = 66 NCH	-	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Textbooks

1. *Samskrta Śāstrom ka Itihās*, Baladev Upadhyaya, 2010, Chowkhambha, Varanasi
2. *A Concise History of Science in India*, D. M. Bose, S. N. Sen and B. V. Subbarayappa, Eds., 22nd Edition, 2010, Universities Press, Hyderabad

Reference Books:

1. Dharampal, *Some Aspects of Earlier Indian Society and Polity and Their Relevance Today, 1987,* New Quest Publications, Pune
2. Dharampal, *Indian Science and Technology in the Eighteenth Century: Some Contemporary European Accounts*, 2021, Dharampal Classics Series, Rashtrotthana Sahitya, Bengaluru
3. Dharampal, *The Beautiful Tree: Indian Indigenous Education in the Eighteenth Century*, 2021, Dharampal Classics Series, Rashtrotthana Sahitya, Bengaluru.
4. J. K. Bajaj and M. D. Srinivas, *Timeless India Resurgent India*, 2001, Centre for Policy Studies, Chennai.

Paper VII/Subject Name: Introduction to Effective Communication		**Subject Code: CEN982A101**
Course Type: AEC		**Course Level: 100**
L-T-P-C – 1-0-0-1	**Credit Units: 01**	**Scheme of Evaluation: T**

Objectives:

The objectives of this course are to make the students understand the four major aspects of communication by closely examining the processes and figuring the most effective ways to communicate with interactive activities.

Prerequisites: None

Course Outcomes:

On successful completion of the course the students will be able to

SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Identify** the elements and processes that make for successful communication and **recognise** everyday activities that deserve closer attention in order to improve communication skills	**BT 1**

CO 2	**Contrast** situations that create barriers to effective communication and **relate** them to methods that are consciously devised to overcome such hindrance	**BT 2**
CO 3	**Use** language, gestures, and para-language effectively to avoid miscommunication and **articulate** one's thoughts and build arguments more effectively	**BT 3**

Detailed Syllabus		
Units	**Course Contents**	**Periods**
I	Introduction to Effective Communication • Listening Skills o The Art of Listening o Factors that affect Listening o Characteristics of Effective Listening • Guidelines for improving Listening skills	**5**
II	• Speaking Skills o The Art of Speaking o Styles of Speaking o Guidelines for improving Speaking skills o Oral Communication: importance, guidelines, and barriers	**5**
III	• Reading Skills o The Art of Reading o Styles of Reading: skimming, surveying, scanning • Guidelines for developing Reading skills	**5**
IV	• Writing Skills o The Art of Writing o Purpose and Clarity in Writing o Principles of Effective Writing	**5**
Total		**20**

Credit Distribution		
Lecture/Tutorial	**Practicum**	**Experiential Learning**
20 hours	-	10 hours - Movie/ Documentary screening - Peer teaching - Seminars - Field Visit

Textbooks:

1. *Business Communication: Essential Starategies for 21st Century Managers,* Shalini Verma, 2nd Edition, 2014, Vikas Publisher

References Books:

1. P.D. Chaturvedi and Mukesh Chaturvedi, *Business Communication,* 4th Edition, 2017, Pearson Education
2. Meenakshi Raman and Sangeeta Sharma, *Technical Communication: Principles and Practice*, 3rd Edition, 2015, Oxford University Press

Paper VIII/Subject Name: Computer Hardware and Networking		Subject Code: CAP052S201
Course Type: SEC		Course Level: 100
L-T-P-C – 3-0-0-3	Credit Units: 03	Scheme of Evaluation: T

Objective:

The objectives of the course are to explain the different hardware components of a computer system and learn its assembling and disassembling along with various networking devices.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** different types of Computers and Networking hardware and software	**BT 1**
CO 2	**Understand** basic idea of installation process and components of a PC	**BT 2**
CO 3	**Experiment** with some hardware components to assemble a computer system or network system.	**BT 3**
CO 4	**Analyse** and **evaluate** different computer systems, network and server for their use.	**BT 4 & 5**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Introduction to Computer Hardware**	Basics of computer organization, Motherboard, CPU, Memory devices (RAM, ROM, Cache), SMPS, Hard disk, SSD, Input/output devices, Ports and connectors, BIOS and CMOS, Peripheral devices, Storage devices and their types.	**15**
II	**PC Assemble and Maintenance**	Computer assembling and disassembling, Installation of operating systems, Device drivers, Formatting and partitioning, Preventive maintenance, Troubleshooting hardware problems, Virus and malware protection, Backup and recovery techniques.	**15**
III	**Fundamentals of Computer Networking**	Introduction to networking, Types of networks (LAN, MAN, WAN), Network topologies, Transmission media, Networking devices: Hub, Switch, Router, Modem, IP addressing, MAC address, OSI and TCP/IP models, Internet and network services.	**15**
IV	**Network Configuration and Security**	Network configuration basics, Wireless networking concepts, Sharing resources in networks, Network troubleshooting tools, Cybersecurity basics, Firewalls, Antivirus software, Data backup and protection, Introduction to cloud networking and emerging networking technologies.	**15**
Total			**60**

Practical Components can include the following:

- Identification of computer hardware components and peripherals.
- Assembling and disassembling a computer system.
- BIOS setup and CMOS configuration.
- Hard disk partitioning and formatting.
- Installation of operating systems and device drivers.
- Installation and configuration of peripheral devices.
- Detection and troubleshooting of hardware faults.

- Preparation and testing of network cables.
- Configuration of peer-to-peer network.
- IP addressing and network configuration.
- Basic router and wireless network configuration.
- Use of networking tools (ping, ipconfig, tracert, etc.).
- Installation and configuration of antivirus/firewall software.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	-	30 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Project)

Text Book:

1. Computer Hardware & Networking With Free CD, A Panel of Authors, 2nd Edition, 2021, Computech Publications

Reference Books:

1. Ajit Mittal and Ajay Rana, *Mastering Pc Hardware And Networking,* 1st Edition, 2014, Khanna Book Publishing.
2. Joginder Singh Saini and Jagdeep Singh Saini, *Royal new pattern computer hardware & network maintenance,* 1st Edition, 2017, Royal Book Dept.

Paper IX/Subject Name: Office Automation		**Subject Code: CAP052S101**
Course Type: VAC		**Course Level: 100**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to familiarize students with modern office automation tools and digital productivity applications, develop practical skills in document creation, spreadsheet management, and presentation design, introduce AI-assisted productivity tools and prompt engineering concepts for workplace automation and enhance digital communication, collaboration, and data handling abilities required in professional environments.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** the concepts and applications of office automation and digital productivity tools.	**BT 2**
CO 2	**Create** professional documents, spreadsheets, and presentations using office software	**BT 3**
CO 3	**Analyse** AI-based productivity tools and prompt engineering techniques for automated content generation and workflow support.	**BT 4**
CO 4	**Evaluate** digital communication and collaboration tools for efficient workplace productivity.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Introduction**	Basics of office automation, computer fundamentals, operating systems, file and folder management, cloud storage concepts,	**05**

		internet and email basics, digital communication tools, office productivity suites (MS Office/LibreOffice/Google Workspace).	
II	**Word Processing**	Creating and formatting documents, tables, templates, mail merge, page formatting, inserting images and charts, document collaboration tools, creating presentations, slide formatting, animations, transitions, presentation best practices.	**07**
III	**Spreadsheet and Data Management**	Spreadsheet basics, formulas and functions, sorting and filtering data, charts and graphs, conditional formatting, pivot tables, basic financial and statistical functions, data validation, worksheet protection, introduction to data analysis and reporting.	**08**
IV	**AI Productivity Tools**	Introduction to AI tools for office productivity, AI-assisted document creation, AI for presentations and spreadsheets, basics of prompt engineering, writing effective prompts, prompt patterns for summarization, email drafting, report generation, data interpretation, ethical use of AI tools, limitations and responsible AI usage.	**10**
Total			**30**

Lab Components based on the following:

- File and folder management operations.
- Creating and formatting professional documents.
- Using tables, templates, and mail merge.
- Designing presentations with multimedia elements.
- Spreadsheet creation and formula implementation.
- Creating charts, graphs, and pivot tables.
- Data analysis using spreadsheet tools.
- Email drafting and professional communication.
- Using AI tools for content generation and summarization.
- Prompt writing for reports, presentations, and spreadsheets.
- Mini project using office automation and AI tools.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 20 NCH = 60 NCH	--	30 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Microsoft Office 365 Complete,* Sandra Cable, Misty Vermaat, and Steven Freund, Cengage Learning.
2. *Computer Fundamentals and Office Automation,* R. K. Taxali, McGraw-Hill Education.
3. *AI Engineering,* Chip Huyen, O'Reilly Media.

Reference Books:

1. C. S. Rayudu, *Business Communication,* Himalaya Publishing House.
2. Michael Alexander and Richard Kusleika, *Data Analysis Using Excel*, Wiley.

SYLLABUS (2nd SEMESTER)

Paper I/Subject Name: Data Structures		**Subject Code: CAP052M201**
Course Type: Core		**Course Level: 100**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to expose the students with the concepts of algorithm design and various types of data structures.

Prerequisites: Basics of C Programming

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** various data structures used in programming.	**BT 1**
CO 2	**Understand** the basic constructs of data structure and its implementation.	**BT 2**
CO 3	**Utilise** the appropriate data structures to solve a given problem.	**BT 3**
CO 4	**Analyse** and **evaluate** the data structures used for problem solving	**BT 4 & 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Linear Data Structure- I**	Introduction: Why we need data structure? Concepts of data structures: Data and data structure, Abstract Data Type and Data Type. Algorithms and programs, basic idea of pseudo-code. Algorithm efficiency and analysis, time and space analysis of algorithms – order notations. Array: Different representations – row major, column major. Sparse matrix - its implementation and usage. Array representation of polynomials. Linked List: Singly linked list, circular linked list, doubly linked list, linked list representation of polynomial and applications.	**15**
II	**Linear Data Structure- II**	**Stack and Queue:** Stack and its implementations (using array, using linked list applications. Queue, circular queue, dequeuers. Implementation of queue- both linear and circular (using array, using linked list), applications. **Recursion:** Principles of recursion – use of stack, differences between recursion and iteration, tail recursion. Applications - The Tower of Hanoi, Eight Queens Puzzle.	**15**
III	**Nonlinear Data Structures**	**Trees:** Basic terminologies, forest, tree representation (using array, using linked list). Binary trees - binary tree traversal (pre-, in-, post- order), threaded binary tree (left, right, full) - non-recursive traversal algorithms using threaded binary tree, expression tree. Binary search tree- operations (creation, insertion, deletion, searching). Height balanced binary tree – AVL tree (insertion, deletion with examples only). **Graphs:** Graph definitions and concepts (directed/undirected graph, weighted/un-weighted edges, sub-graph, degree, cut-vertex/articulation point, pendant node, clique, complete graph,	**15**

		connected components – strongly connected component, weakly connected component, path, shortest path, isomorphism). Graph representations/storage implementations – adjacency matrix, adjacency list, adjacency multi-list. Graph traversal and connectivity – Depth-first search (DFS), Breadth-first search (BFS) – concepts of edges used in DFS and BFS (tree-edge, back-edge, cross-edge, and forward-edge), applications. Minimal spanning tree – Prim's algorithm (basic idea of greedy methods). B-Trees operation	
IV	**Searching, Sorting**	**a. Sorting Algorithms:** Bubble sort and its optimizations, insertion sort, shell sort, selection sort, merge sort, quick sort, heap sort (concept of max heap, application – priority queue), radix sort. **b**. **Searching Algorithms:** Sequential search, binary search, interpolation search.	**15**
Total			**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 15 NCH = 45 NCH	--	15 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Data structures, Algorithms and Applications in C++*, S. Sahni, 2nd Edition, 2004, University Press (India) Pvt. Ltd.
2. *Data structures and Algorithms in C++,* Michael T. Goodrich, R. Tamassia and. Mount, 2nd Edition, 2011, John Wiley and Sons

Reference Books:

1. Seymour Lipschutz, *Data Structures*, 1st Edition (reprint) 2017, McGraw Hill Education.
2. Yashavant P. Kanetkar, *Data Structure through C++*, 2nd Edition, 2003, BPB Publications.

Paper II/Subject Name: Computer Architecture		**Subject Code: CAP052M202**
Course Type: Core		**Course Level:100**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand the machine instruction, basic computer organization and memory hierarchy with pipelining processing.

Prerequisites: Basics of Digital Logic and Computer Design

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** & **explain** the functional units and operational principles of computer systems.	**BT 1 & 2**
CO 2	**Apply** data representation and instruction formats in computer architecture problems.	**BT 3**

CO 3	**Analyse** CPU organization, memory hierarchy, and input/output mechanisms.	**BT 4**
CO 4	**Evaluate** the performance of computer systems using architectural concepts and techniques.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Content	Periods
I	**Introduction**	Basic structure of computers, Functional units, Von Neumann architecture, Instruction execution cycle, Bus structure, Data representation, Fixed-point and floating-point representation, Binary arithmetic, Character representation codes (ASCII, Unicode).	**15**
II	**Processor Organisation and Instruction Set**	CPU organization, Registers and register transfer language, Instruction formats, Addressing modes, Instruction cycle, Stack organization, Arithmetic and logic unit (ALU), Control unit: hardwired and microprogrammed control.	**15**
III	**Memory Organisation**	Memory hierarchy, Primary memory: RAM, ROM, Cache memory, Secondary storage devices, Associative memory, Virtual memory concepts, Cache mapping techniques, Memory management basics.	**15**
IV	**I/O Organisation**	I/O interface, Modes of data transfer, Interrupts, DMA, Pipelines and pipelining concepts, Parallel processing basics, RISC and CISC architecture, Overview of multicore processors and modern computing architectures.	**15**
Total			**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	-	30 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Project)

Text Book:

1. *Computer System and Architecture*, Moris Mano, 3rd Edition, 2007, PHI.
2. *Structured Computer Organization,* A. S. Tanenbaum, 5th Edition, 2009, Prentice Hall of India

Reference Books:

1. V. C. Hamacher, Z. G. Vranesic and S. G. Zaky, *Computer Organization,* 5th Edition, 2002 McGraw Hill.
2. J. L. Hennessy and D. A. Patterson, *Computer Architecture: A Quantitative Approach,* 4/e, 2006, Morgan Kaufmann.
3. D. V. Hall, *Microprocessors and Interfacing,* 2nd Edition, 2006, McGraw Hall.

Paper III/Subject Name: OOP using C++		**Subject Code:**
Course Type: Core		**Course Level: 100**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students familiar with the principles and concepts of OOP, develop programming skills using C++ language constructs and object-oriented features, enable students to design modular,

reusable, and efficient software solutions and familiarize students with advanced OOP concepts such as inheritance, polymorphism, and file handling.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** and **explain** the principles of object-oriented programming and C++ program structure.	**BT 1 & 2**
CO 2	**Develop** C++ programs using classes, objects, constructors, and member functions.	**BT 3**
CO 3	**Analyze** program requirements and implement appropriate OOP concepts for problem-solving.	**BT 4**
CO 4	**Evaluate** and debug C++ programs involving files, exceptions, and dynamic memory management	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Introduction**	Introduction to Object-Oriented Programming, Features of OOP, Structure of C++ program, Tokens, Data types, Variables, Operators, Control structures, Functions, Inline functions, Default arguments, Function overloading, Introduction to classes and objects.	**15**
II	**Classes and Objects**	Class definition, Access specifiers, Member functions, Constructors and destructors, Types of constructors, Copy constructor, Friend functions, Static data members and functions, Arrays of objects, Objects as function arguments, Operator overloading.	**15**
III	**Inheritance and Polymorphism**	Inheritance: single, multilevel, multiple, hierarchical and hybrid inheritance, Visibility modes, Virtual base class, Polymorphism, Function overriding, Virtual functions, Abstract classes, Runtime polymorphism, Templates and generic programming basics.	**15**
IV	**Advanced Concepts**	Exception handling, File handling in C++, Streams, Dynamic memory allocation using new and delete, Pointers and objects, Command line arguments, Introduction to Standard Template Library (STL), Applications of OOP in software development.	**15**
Total			**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	-	30 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Project)

Text Book:

1. *Object Oriented Programming with C++,* E. Balagurusamy, 8th Edition, McGraw-Hill Education.
2. *The C++ Programming Language,* Bjarne Stroustrup, 4th Edition, Addison-Wesley.

Reference Books:

1. Robert Lafore, *Object-Oriented Programming in C++*, Sams Publishing.
2. Stanley B. Lippman, Josée Lajoie, and Barbara Moo, *C++ Primer*, Addison-Wesley.

Paper IV/Subject Name: Data Structures Lab		**Subject Code: CAP052M201**
Course Type: Core		**Course Level: 100**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to expose the students with the concepts of algorithm design and various types of data structures.

Prerequisites: Basics of C Programming

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** various data structures used in programming.	**BT 1**
CO 2	**Understand** the basic constructs of data structure and its implementation.	**BT 2**
CO 3	**Utilise** the appropriate data structures to solve a given problem.	**BT 3**
CO 4	**Analyse** and **evaluate** the data structures used for problem solving	**BT 4 & 5**

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 20 Laboratory experiments based on the following:

1. Some common programs of C as revision.
2. Programs on Arrays- Traversal, Insertion, Deletion, Polynomial Representation, etc.
3. Programs on Linked List- Creation Insertion, Deletion, Polynomial Representation, etc.
4. Programs on Stacks-Creation, Push Pop, Infix to Postfix Conversion, Evaluation.
5. Programs on Queues-Creation, Insertion, Deletion, etc.
6. Programs ‘
7. on Trees- Binary Tree Creation, Tree Traversal, BST
8. Programs on Searching- Linear Search, Binary Search
9. Programs on Sorting- Bubble Sort, Insertion Sort, Selection Sort, Quick Sort, Merge Sort, Heap Sort.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
	20 NCH	10 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Data structures, Algorithms and Applications in C++*, S.Sahni, 2nd Edition, 2004, University Press (India) Pvt. Ltd.
2. *Data structures and Algorithms in C++,* Michael T.Goodrich, R.Tamassia and .Mount, 2nd Edition, 2011, John Wiley and Sons

Reference Books:

1. Seymour Lipschutz, *Data Structures*, 1st Edition (reprint) 2017, McGraw Hill Education.
2. Yashavant P. Kanetkar, *Data Structure through C++*, 2nd Edition, 2003, BPB Publications.

Paper V/Subject Name: OOP using C++Lab		**Subject Code:**
Course Type: Core		**Course Level: 100**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to make the students familiar with the principles and concepts of OOP, develop programming skills using C++ language constructs and object-oriented features, enable students to design modular, reusable, and efficient software solutions and familiarize students with advanced OOP concepts such as inheritance, polymorphism, and file handling.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** and **explain** the principles of object-oriented programming and C++ program structure.	**BT 1 & 2**
CO 2	**Develop** C++ programs using classes, objects, constructors, and member functions.	**BT 3**
CO 3	**Analyze** program requirements and implement appropriate OOP concepts for problem-solving.	**BT 4**
CO 4	**Evaluate** and debug C++ programs involving files, exceptions, and dynamic memory management	**BT 5**

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 20 Laboratory experiments based on the following:

1. Study of C++ program structure and basic input/output operations.
2. Programs using control statements, functions, and function overloading.
3. Programs using classes and objects.
4. Programs demonstrating constructors and destructors.
5. Programs using friend functions and static members.
6. Programs for operator overloading.
7. Programs demonstrating single and multilevel inheritance.
8. Programs demonstrating multiple and hierarchical inheritance.
9. Programs implementing function overriding and virtual functions.
10. Programs demonstrating runtime polymorphism.
11. Programs using templates and generic functions.
12. Programs demonstrating dynamic memory allocation using new and delete.
13. Programs using exception handling mechanisms.
14. File handling operations in C++ (text and binary files).
15. Mini project using object-oriented concepts.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	20 NCH	10 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Project)

Text Book:

1. *Object Oriented Programming with C++,* E. Balagurusamy, 8th Edition, McGraw-Hill Education.
2. *The C++ Programming Language,* Bjarne Stroustrup, 4th Edition, Addison-Wesley.

Reference Books:

1. Robert Lafore, *Object-Oriented Programming in C++*, Sams Publishing.
2. Stanley B. Lippman, Josée Lajoie, and Barbara Moo, *C++ Primer*, Addison-Wesley.

Paper VI/Subject Name: Introduction to Indian Knowledge System-II		**Subject Code: IKS982I201**
Course Type: IKS		**Course Level: 100**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objectives:
The objective of this course id to present an overall introduction to all the streams of IKS relevant to the UG programme. It would enable students to explore the most fundamental ideas that have shaped Indian Knowledge Traditions over the centuries.

Prerequisite: Basic knowledge of IKS-I

Course Outcomes:
On completion of this course students will be expected to –

CO	Contents	BT Level
CO1	**Recall** about classical literature in Sanskrit and other languages	**BT 1**
CO2	**Recall** traditional Indian knowledge system and Indian education	**BT 1**
CO3	**Summarize the Indian** Art, Architecture, Agriculture, Polity and Economy	**BT 2**

Detailed Syllabus:

Module	Course Contents	Periods
I	**Classical Literature in Sanskrit and Other Indian Languages**: The nature and purpose of Kavya. Drisya and Sravya Kavyas. The ideas of Indian aestheticians on what constitutes the soul of Kavya. Important examples of classical literature in Sanskrit and other Indian languages **Indian Education**: Preservation of culture, tradition and Dharma through education. Svadhyaya, Pravachana. Also continuity of the family and the vamsha, who are the carriers of knowledge, tradition and Dharma. The extent, inclusiveness and the sophistication of indigenous education in early19th century India. **The Purpose of Knowledge in India**: Para Vidya and Apara Vidya. The corpus connected with Para Vidya. Learning and formalization of concepts associated with Para Vidya also form part of Apara. Apara Vidya. Nature and purpose of sciences, technologies, and all human knowledge concerning the world and the society. The concept of Rita, Dharma. The cycle of mutual dependence of humans and all aspect of creation. Yajna and the inviolable discipline of sharing and caring.	**15**

II	**Methodology of Indian Knowledge System**: Systematization of knowledge fields as Sastra. Each Sastra has a clearly defined purpose in Vyavahara. The means of valid knowledge (Pramanas). Perception (Pratyaksha), Inference (Anumana) and Textual Tradition (Agama), as discussed in the canonical texts of all the disciplines. The importance of Pratyaksha and Agama in relation to Anumana. **Indian Architecture and Town Planning**: The importance of Sthapatya-veda. The ancient cities of the Indus Saraswati region. Town planning and drainage systems. Examples of the significance of architecture and materials in Ramayana and Mahabharata. Public opulence and private austerity in Indian architecture. Why there are many more of Temples than Palaces. Important texts of Architecture and Sculpture. The prevalence of high Indian architecture in almost all parts of India except the Ganga plains. Examples of high Indian architecture from ancient and medieval periods from different parts of India. The building of Jaipur in the 18th century. How temple art and architecture continue to flourish in modern India. **Indian Fine Arts:** The importance of Gandharva-veda. Natyasastra on the nature and purpose of fine arts. Basic concepts of Indian music and dance. Important texts of Indian music, dance and painting. Indian musical instruments. Different schools of music, dance and painting in different regions of India. Important examples of Indian painting in various part of India. Musicology as a science. Harmonising Lakshya and Lakshana (practise and theory). Major developments in the science and practice of music the 17/18/19th centuries. The current revival of music and dance in India.	**15**
III	**Indian Agriculture**: The significance of agriculture and irrigation as emphasised in the Ramayana, Mahabharata and other texts. Mention of Indian agriculture by the Greek historians and later travellers. Significance of agriculture and irrigation for the kings of Indian tradition. Major water-bodies of the ancient times. The Ery system of south India. Excellence of Indian agricultural technologies as observed by more recent European observers. Productivity of Indian agriculture in medieval Thanjavur and eighteenth century Allahabad, Chengalpattu, etc. Indian attitude towards agriculture, based on Walker and later reports. **Indian Textiles**: India as the ancient home of cotton and silk fabrics. Weaving formed the most significant part of Indian economy after agriculture. Varieties of textiles and dyes developed in different regions of India. India as a leading exporter of textiles in the world in the 17/18/19th centuries. **Indian Metallurgy:** Vedic references to metals and metal working. Mining and manufacture in India of Zinc, Iron, Copper, Gold, etc., from ancient times. Indian texts which refer to metallurgy. Important specimens of metal workmanship preserved/found in different parts of India. The significance and wide prevalence of ironsmith and other metal workers in the pre-modern era. European observers on the high quality and quantity of Indian iron and steel in the 18/19th centuries.	**15**
IV	**Indian Polity and Economy**: Indian conception of well-organised Polity and flourishing Economy as expounded in the foundational texts. The notion of Bharartavarsha as a Chakravarti-Kshetra and important attributes of Chakravartin. King as the protector of Dharma. King as the strength and support of the weak. King as the protector of Varta. King as the protector of the times. Meaning of Varta: Krishi, Gopalana and Vanijya forming the basis of Varta and the core of economic activity in society. The importance of sharing. Grama as the centre of the polity. **The Outreach of Indian Knowledge System**: The outreach of Indian Knowledge System beyond Indian boundaries forms the ancient times. Outreach to East, Southeast, Central and Southeast Asia of Indian phonetic script, decimal value place system-based arithmetic, algebra, astronomy and calendar, medical pharmacopeia, architecture, methods of making iron and steel, cotton textiles, etc. The transmission of Indian linguistics, knowledge of plants, iron and steel metallurgy, textiles and dyeing, shipbuilding etc., to Europe in 17/18/19th centuries. Current global outreach of Ayurveda, Yoga and Indian Fine Arts.	**15**
	Total	**60**

Credit Distribution

Lecture/ Tutorial	Practicum	Experiential Learning
60 NCH	-	30 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Textbooks:

1. *Samskrta Śāstrom ka Itihās,* Baladev Upadhyaya, 2010, Chowkhambha, Varanasi.
2. *A Concise History of Science in India,* D. M. Bose, S. N. Sen and B. V. Subbarayappa, Eds., 2nd Edition, 2010, Universities Press, Hyderabad

Reference Books:

1. Astāngahrdaya, Vol. I, Sūtrasthāna and Śarīrasthāna, Translated by K. R. Srikantha Murthy, Vol. I, Krishnadas Academy, Varanasi, 1991.
2. Dharampal, Some Aspects of Earlier Indian Society and Polity and Their Relevance Today, New Quest Publications, Pune, 1987.
3. Dharampal, Indian Science and Technology in the Eighteenth Century: Some Contemporary European Accounts, Dharampal Classics Series, Rashtrotthana Sahitya, Bengaluru, 2021
4. Dharampal, The Beautiful Tree: Indian Indigenous Education in the Eighteenth Century, Dharampal Classics Series, Rashtrotthana Sahitya, Bengaluru, 2021.
5. J. K. Bajaj and M. D. Srinivas, Indian Economy and Polity in Eighteenth century Chengalpattu, in J. K. Bajaj ed., Indian Economy and Polity, Centre for Policy Studies, Chennai, 1995, pp. 63-84.
6. J. K. Bajaj and M. D. Srinivas, Annam Bahu Kurvita Recollecting the Indian Discipline of Growing and Sharing Food in Plenty, Centre for Policy Studies, Chennai, 1996.
7. J. K. Bajaj and M. D. Srinivas, Timeless India Resurgent India, Centre for Policy Studies, Chennai, 2001.
8. M. D. Srinivas, The methodology of Indian sciences as expounded in the disciplines of Nyāya, Vyākarana, Ganita and Jyotisa, in K. Gopinath and Shailaja D. Sharma (eds.), The Computation Meme: Explorations in Indic Computational Thinking, Indian Institute of Science, Bengaluru, 2022 (in press).

Paper VII/Subject Name: Approaches to Verbal and Non-Verbal Communication		**Subject Code: CEN982A201**
Course Type: AEC		**Course Level: 100**
L-T-P-C – 1-0-0-1	**Credit Units: 01**	**Scheme of Evaluation: T**

Course Objectives

To introduce the students to the various forms of technical communication and enhance their knowledge in the application of both verbal and non-verbal skills in communicative processes.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Identify** the different types of technical communication, their characteristics, their advantages and disadvantages.	**BT 1**
CO 2	**Explain** the barriers to communication and ways to overcome them.	**BT 2**
CO 3	**Discover** the means to enhance conversation skills.	**BT 3**
CO 4	**Determine** the different types of non-verbal communication and their significance.	**BT 3**

Detailed Syllabus

Modules	Topics (if applicable) & Course Contents	Periods
I	**Technical Communication** Communicating about technical or specialized topics, Different forms of technology-enabled communication tools used in organisations Telephone, Teleconferencing, Fax, Email, Instant messaging, Blog, podcast, Videos, videoconferencing, social media	**5**
II	**Communication Barriers** Types of barriers: Semantic, Psychological, Organisational, Cultural, Physical, and Physiological. Methods to overcome barriers to communication.	**5**
III	**Conversation skills/Verbal Communication** Conversation – Types of Conversation, Strategies for Effectiveness, Conversation Practice, Persuasive Functions in Conversation, Telephonic Conversation and Etiquette Dialogue Writing, Conversation Control.	**5**
IV	**Non-verbal Communication** Introduction; Body language- Personal Appearance, Postures, Gestures, Eye Contact, Facial expressions Paralinguistic Features-Rate, Pause, Volume, Pitch/Intonation/ Voice/ modulation Proxemics , Haptics, Artifactics, Chronemics	**5**
	Total	**20**

Credit Distribution		
Lecture/Tutorial	**Practicum**	**Experiential Learning**
20 hours	-	10 hours - Movie/ Documentary screening - Peer teaching - Seminars - Field Visit

Textbooks:

1. Rizvi, M. Ashraf. *Effective Technical Communication*. 2017. McGraw-Hill.
2. Chaturvedi, P. D. and Chaturvedi, Mukesh. *Business Communication.* 2014. Pearson.
3. Raman, Meenakshi and Sharma, Sangeeta. *Technical Communication: Principles and Practice*, 2nd Edition. 2011. Oxford University Press.

Reference Books:

1. Hair, Dan O., Rubenstein, Hannah and Stewart, Rob. *A Pocket Guide to Public Speaking.* 5th edition, 2015. St. Martin's. ISBN-13:978-1457670404
2. Koneru, Aruna. *Professional Communication*. 2017. Tata McGraw Hill ISBN-13: 978-0070660021
3. Raman, Meenakshi and Singh, Prakash. *Business Communication*. 2nd Edition. 2012. Oxford University Press
4. Sengupta, Sailesh. *Business and Managerial Communication*. 2011. PHI Learning Pvt. Ltd.

Paper VIII/Subject Name: Internet Technologies		**Subject Code: CAP052S201**
Course Type: SEC		**Course Level: 100**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students learn the fundamentals of internet technologies and web architecture and client- server-side web technologies, and to develop skills in designing static and dynamic web pages using HTML, CSS, JavaScript, and PHP.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** and **explain** internet technologies, web architecture, and web development concepts**.**	**BT 1 & 2**
CO 2	**Design** and **develop** responsive web pages using HTML, CSS, and JavaScript and PHP.	**BT 3**
CO 3	**Integrate** databases with PHP applications for data storage and retrieval.	**BT 4**
CO 4	**Evaluate** web applications for functionality, usability, and security.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Introduction**	Evolution of Internet, Internet services, TCP/IP basics, Web browsers and web servers, URL, DNS, HTTP and HTTPS, Introduction to web development, Structure of HTML documents, HTML elements and attributes, Headings, paragraphs, lists, hyperlinks, images, tables, forms, multimedia elements, semantic HTML5 tags.	**15**
II	**CSS and JavaScript**	Introduction to CSS, Inline, internal and external style sheets, Selectors, colors, fonts, box model, positioning, responsive design basics, Introduction to JavaScript, Variables, operators, control statements, functions, arrays, events, DOM manipulation, form validation.	**15**
III	**PHP Programming**	Introduction to PHP, PHP syntax, Variables and data types, Operators and expressions, Control statements, Functions, Arrays, String handling, Form handling using GET and POST methods, Sessions and cookies, File handling, Error handling basics.	**15**
IV	**PHP with Database Connectivity**	Introduction to MySQL, Database creation and tables, SQL queries, PHP-MySQL connectivity, CRUD operations, User authentication, Simple login and registration systems, Introduction to web security concepts, Deployment basics and overview of modern web frameworks.	**15**
Total			**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	-	30 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Project)

Text Book:

1. *Internet and World Wide Web How to program*,Deitel H.M. and Deitel P.J, 4th Edition, 2012, Pearson International, New Delhi.
2. *Web Technology,*Gopalan N.P. and Akilandeswari J., 2nd Edition, 2014. Prentice Hall of India.
3. *Java How to Program*, Paul Dietel and Harvey Deitel, 8th Edition, 2014, Prentice Hall of India.

Reference Books:

1. Uttam K.Roy, *Web Technologies*, 2011, Oxford University Press.
2. Godbole A. S. & Kahate A., TMH, *Web Technologies*, 2ndEdition, 2006, TMH

Paper IX/Subject Name: Introduction to Data Analytics		**Subject Code:**
Course Type: VAC		**Course Level: 100**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to make the students learn the fundamentals of data analytics and business intelligence concepts, develop skills in data visualization and dashboard creation using Power BI, enable students to perform data transformation, analysis, and reporting and familiarize students with data-driven decision-making techniques and analytics workflows.

Prerequisites: Basic knowledge of computer applications, spreadsheets, and statistics fundamentals.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** the concepts of data analytics, business intelligence, and data visualization.	**BT 2**
CO 2	**Import** clean, and transform data using Power BI tools and create interactive dashboards and visual reports for data analysis.	**BT 3**
CO 3	**Analyze** datasets using Power BI functions and visualization techniques	**BT 4**
CO 4	**Evaluate** data insights and prepare analytical reports for decision-making.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Introduction**	Introduction to data analytics, types of analytics (descriptive, diagnostic, predictive, prescriptive), Business Intelligence concepts, Data sources and data types, Introduction to Power BI interface, Installing and configuring Power BI Desktop, Data analytics workflow, Applications of analytics in business and industry.	**05**
II	**Data Preparation and Transformation**	Importing data from Excel, CSV, databases, and web sources, Data cleaning techniques, Handling missing and duplicate data, Data transformation using Power Query Editor, Data modeling basics, Relationships between tables, Data types and formatting.	**07**
III	**Data Visualization and Dashboard Design**	Creating charts and visualizations, Tables, matrices, bar charts, pie charts, line charts, maps, slicers, filters, KPIs, Drill-down and drill-through features, Dashboard design principles, Interactive reports, Formatting and customization of reports.	**08**

IV	**Data Analysis and Reporting using Power BI**	Introduction to DAX (Data Analysis Expressions), Calculated columns and measures, Basic analytical functions, Time intelligence basics, Report publishing and sharing, Generating insights from dashboards, Introduction to AI visuals in Power BI, Ethics and security in data analytics.	**10**
Total			**30**

Lab Components based on the following:

- Installation and familiarization with Power BI Desktop
- Importing data from Excel and CSV files
- Data cleaning and transformation using Power Query
- Creating relationships between datasets
- Creating basic charts and visualizations
- Applying filters, slicers, and drill-down features
- Creating interactive dashboards
- Using DAX for calculated columns and measures
- Generating analytical reports and insights
- Mini project using real-world datasets

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 20 NCH = 60 NCH	--	30 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Beginning Power BI,* Dan Clark, Apress.
2. *Microsoft Power BI Cookbook,* Brett Powell, Microsoft Power BI Cookbook, Packt Publishing.
3. *Data Analytics Made Accessible,* Anil Maheshwari, Data Analytics Made Accessible, CreateSpace Independent Publishing.

Reference Books:

1. Cole Nussbaumer Knaflic, *Storytelling with Data: A Data Visualization Guide for Business Professionals*, Wiley.
2. Devinkumar Chanapathi, *Practical Business Intelligence with Power BI*, BPB Publications.
3. Claus O. Wilke, *Fundamentals of Data Visualization*, O'Reilly Media.